GOBIERNO DE CHILE
MINISTERIO DE ECONOMÍA, FOMENTO Y TURISMO

# RfP 2 Predictive Outlook: What This Means for Tomorrow

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| **To** | Office of the Vice Minister of Tourism, Ministerio de Economía, Fomento y Turismo |
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| **Analytical frame** | 15-region legacy dataset; DB3 quarterly panel 2019-Q1–2025-Q3, DB4 Digital Signals, DB5 readiness; builds on the RfP 1 tiers |
| **Date** | September 2026 |

## Forecast conclusion:

The predictive evidence supports action. We recommend quarterly forecasting as an early-warning tool for investment and operating decisions, with human review before budget, access, permitting or safety actions are triggered. Two priority-region forecasts were tested on data the model never saw (2024-Q4 to 2025-Q3) and both beat the required same-quarter-last-year baseline. The same specification, however, loses to that baseline in 26 of 45 region–metric tests, so we recommend it as a governed challenger for specific series rather than replacement of the junior analyst's rule.

| Region | Metric | Training / test | Model RMSE | Naive RMSE | Improvement | Confidence |
|---|---|---|---|---|---|---|
| Coquimbo | International arrivals | 11 quarters / 4 quarters | 1,506 | 2,259 | 33% lower RMSE | Moderate-high; one of only two arrivals series where the model wins |
| Valparaíso | Occupancy rate (points) | 11 quarters / 4 quarters | 10.85 | 13.76 | 21% lower RMSE | Moderate; still 16% better without the 2025-Q1 outlier (3.2 vs 3.9) |

The model is simple: an intercept, a linear time trend and quarterly seasonal effects, five parameters fitted to the eleven post-shock quarters from 2022-Q1 to 2024-Q3 (six residual degrees of freedom), with international arrivals modelled on a log scale. It is preferable to a more complicated model that cannot be defended with such few observations; the goal is decision reliability. For the outlook, the model is refit through 2025-Q3 and shown against a seasonal-naive planning baseline (Figure 3).

**Figure 3. Seasonal-trend model vs seasonal-naive baseline: 4-quarter holdout test and 8-quarter outlook**




| Outlook 2025-Q4 → 2027-Q3 | 2025-Q4 | 2026-Q1 | 2026-Q2 | 2026-Q3 | 2026-Q4 | 2027-Q1 | 2027-Q2 | 2027-Q3 |
|---|---|---|---|---|---|---|---|---|
| **Coquimbo intl arrivals: model** | 14,752 | 19,052 | 12,948 | 10,418 | 17,865 | 23,073 | 15,680 | 12,617 |
| Coquimbo: seasonal-naive floor | 12,639 | 16,060 | 11,184 | 6,981 | 12,639 | 16,060 | 11,184 | 6,981 |
| **Valparaíso occupancy %: model** | 67.5 | 76.2 | 65.5 | 60.1 | 71.5 | 80.1 | 69.4 | 64.1 |
| Valparaíso: seasonal-naive floor | 62.8 | 86.2 | 56.7 | 51.6 | 62.8 | 86.2 | 56.7 | 51.6 |

Read the two lines as a planning range, not as exact predictions. The refit Coquimbo trend compounds at about +4.9% per quarter, so by 2027-Q1 the model sits 44% above the naive floor (23,073 vs 16,060 arrivals), that slope is the post-pandemic recovery, and recoveries decelerate. The Valparaíso trend adds about 3.9 occupancy points a year, which cannot continue indefinitely. We would therefore budget staffing, mobility and accommodation capacity to the floor, stress-test against the model and re-estimate quarterly resilience and sustainable development, not visitor volume alone (World Economic Forum, 2024).

## How far can the model be trusted? (The same test on every region and metric)

| Scale Now region | International arrivals | Occupancy rate | Average daily rate |
|---|---|---|---|
| Metropolitana | −42% | −89% | +30% |

| Scale Now region | International arrivals | Occupancy rate | Average daily rate |
|---|---|---|---|
| Valparaíso | −239% | **+21%** | −2% |
| Coquimbo | **+33%** | −220% | **+47%** |
| Los Lagos | −274% | −67% | −80% |
| Magallanes y Antártica | −245% | **+17%** | −125% |

Cells show the RMSE reduction of the model against the seasonal-naive rule on the same 2024-Q4 to 2025-Q3 holdout (positive = model better). Across all 45 region-metric series the model wins 19 (42%): only 2 of 15 arrivals series, 6 of 15 occupancy series and 11 of 15 average-daily-rate series, where price trends are steadier. For arrivals the log-linear trend usually over-extrapolates the recovery slope, and the junior analyst's rule is the better default; Coquimbo is the exception, not the pattern. Our recommendation follows: run the model as a challenger against the naive champion for every series, promote it only where it has beaten the baseline for two consecutive holdout windows, and demote it when it stops. The full 45-test matrix is in the supporting workbook (Forecast Generalization) and in Dashboard 04. Read a negative percentage as how much larger the model's error is than the baseline's, not as a share of 100: Valparaíso international arrivals comes in at −239%, meaning the model's holdout error (RMSE 6,013) was more than three times the size of simply assuming this quarter matches the same quarter last year (naive RMSE 1,773), a clear loss and exactly the kind of series the champion/challenger rule above is built to catch and demote.

## What the semi-structured data changes?

DB4 provides information that the historical spreadsheets cannot provide. It offers near-real-time visitor pressure, digital sentiment, platform behaviour and origin-destination movement. It contains 1,001 illustrative records across booking-platform snapshots, social-media sentiment, IoT sensors and mobile-flow estimates, each with its own fields and gaps. As coverage is uneven, these signals modify decisions rather than replacing the core evidence and each row below states what it changes.

| Signal | Specific evidence | What it changes |
|---|---|---|
| Chiloé protected-area pressure | Parque Nacional Chiloé trail counters average 749.5 visitors per reading, peak 891 (n = 6 readings); 2 of 6 (33%) show non-open status. | Los Lagos stays Scale Now, but nature promotion is coupled with timed entry, dispersal, maintenance and local capacity funding (D2 is its largest budget line). |
| Rapa Nui operational pressure | Parque Nacional Rapa Nui trail counters average 684.9, peak 770 (n = 7 readings); 4 of 7 (57%) are partial or weather closures. | Valparaíso stays Scale Now; the island component is capacity-managed rather than volume-maximised. |
| Sentiment divergence | 2025 mention-weighted sentiment: +0.69 Metropolitana and +0.68 Magallanes (up from 0.26 in 2024) versus -0.05 Tarapacá, -0.02 Araucanía and 0.00 Maule (down from +0.48). | Digital promotion goes first to the positive pair; the negative trio gets service, mobility and safety diagnosis before paid reach, Maule's slide is a watch item for its Prepare First plan. |
| Corridor opportunity | Mobile-flow estimates: Metropolitana → Coquimbo 11,675 visitors in 2025-Q2 with 3.7 average nights; Maule → Los Lagos 11,695 with 2.9 nights. | Build corridor bundles and short-break packages; market Coquimbo as a Santiago-linked astro-coastal corridor instead of region-isolated campaigns. |
| Complaint themes (new) | Across 2024–25 flagged complaints the top tags are overcrowding at peak season (11), price transparency (11), signage (10) and cleanliness near terminals (10). | Crowding is already a visitor-experience problem, which supports capacity management before more promotion; signage and cleanliness are cheap Prepare First fixes. |
| Sensor reliability (new) | 43 of 260 IoT readings lack battery data and 36 report battery below 20%; 28 of 80 trail-counter readings are closures. | Any monitoring investment must fund sensor upkeep, or gaps will be misread as low visitation; this is written into the cross-cutting budget line. |

Visa's required reading supports these findings: planning and booking have gone digital, 58% of travellers book trip elements separately and affluent Latin American travellers rely on review sites. So, corridor bundles must be modular and bookable online. Eco-friendly options are also preferred when they cost the same (86% of Mexican travellers). This favours capacity-managed nature tourism over undifferentiated growth (Visa, 2023, pp. 7-17). The vignette's warning that data gaps concentrate in informal tourism and unregistered accommodation is the likely finding of the platform-versus-official price gap in Tarapacá, O'Higgins and Maule (Serrato, n.d.).

## Two prediction-to-action chains and the cost of being wrong

| Prediction | Data in | Human judgment | Action | Cost of being wrong | Success / retraining trigger |
|---|---|---|---|---|---|
| Crowding risk at a protected area (Chiloé, Rapa Nui) next quarter | IoT trail counters and status; mobile inflows; DB3 occupancy; weather; social complaints. | Rangers validate whether closures reflect crowding, maintenance or weather; local authorities assess community impacts. | Timed-entry thresholds, shuttle capacity, ranger staffing, maintenance windows, dispersal and promotion throttling. | False alarm: lost revenue and operator frustration from unnecessary throttling. Missed surge: safety incidents, trail damage, closures, reputational harm, the asymmetry favours acting on moderate confidence. | Fewer emergency closures and complaints without suppressing spend. Retrain when forecast error or the sensor–behaviour link deteriorates for two consecutive quarters. |
| Accommodation capacity pressure in Valparaíso or Coquimbo within 12-18 months | DB3 occupancy and ADR; platform occupancy, rates and reviews; arrivals; mobile flows; sentiment. | Officials separate short-lived events from structural demand; investors assess the pipeline and neighbourhood constraints. | Lodging incentives or faster permitting where capacity is structural; otherwise pricing and transport management instead of overbuilding. | False positive: subsidised overbuilding, falling occupancy, stranded assets. False negative: price spikes, service failures, leakage to informal supply, costs on both sides, so act in stages. | Occupancy stays healthy without persistent price spikes or service failures. Retrain when RMSE rises >25% over the validation benchmark or a structural break occurs. |

## Should Chile invest in predictive, AI-enabled monitoring now?

Yes, but it should be used as a decision support tool with proper oversight, even though current predictions are not perfect. Earlier visibility is worth a lot in tourism because capacity, safety, conservation and promotion decisions are time-sensitive, and the signals

already exist across bookings, sensors, mobile flows and social channels. The course vignette describes exactly this division of labour: analytics can scan large indicator sets, flag anomalies and summarise patterns, but descriptive relationships still require qualitative investigation and must not be mistaken for causal proof (Serrato, n.d.). Our own generalisation test is the case in point, the same model is trustworthy for some series and wrong for others, which argues for building the validation routine, not skipping it. We therefore fund it: USD 1.5 million is ring-fenced for sensor upkeep, a shared data platform, a privacy and model-risk audit and regional data stewards.

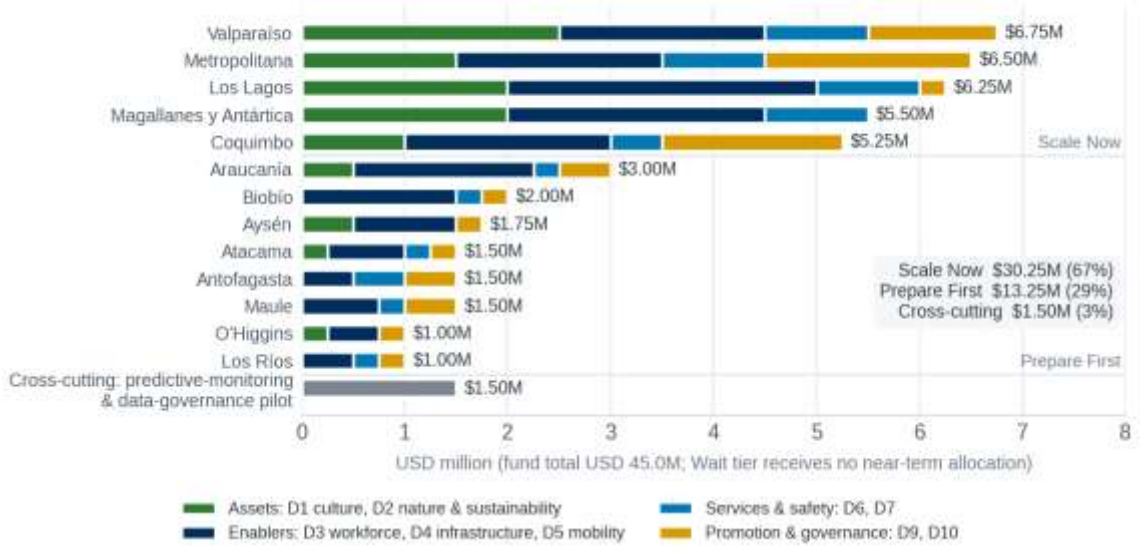
The architecture is 'human-in-command': models detect and forecast; accountable officials decide. Mobile-flow data is accepted only in aggregate, with minimum cell-size thresholds, retention limits, purpose limitation, access logging, vendor audit rights and a ban on re-identification attempts. The Ministry should publish a plain-language data-use statement and an annual model-risk report; if those safeguards cannot be independently verified, mobile-flow data should be excluded even at the cost of accuracy, the corridor insights are useful, but not at the price of citizens' trust.

## USD 45 million allocation

| Region | Tier | Allocation | Largest lines |
|---|---|---|---|
| Metropolitana | Scale Now | **$6.50M** | D1 $1.50M, D9 $1.50M, D5 $1.00M |
| Valparaíso | Scale Now | **$6.75M** | D2 $1.50M, D1 $1.00M, D4 $1.00M |
| Coquimbo | Scale Now | **$5.25M** | D9 $1.50M, D5 $1.00M, D2 $1.00M |
| Los Lagos | Scale Now | **$6.25M** | D2 $2.00M, D4 $1.00M, D3 $1.00M |
| Magallanes y Antártica | Scale Now | **$5.50M** | D2 $2.00M, D5 $1.00M, D4 $1.00M |
| Maule | Prepare First | **$1.50M** | D3 $0.50M, D10 $0.25M, D4 $0.25M |
| Biobío | Prepare First | **$2.00M** | D3 $0.50M, D5 $0.50M, D4 $0.50M |
| Antofagasta | Prepare First | **$1.50M** | D5 $0.25M, D9 $0.25M, D4 $0.25M |
| Araucanía | Prepare First | **$3.00M** | D3 $0.75M, D10 $0.50M, D2 $0.50M |
| Atacama | Prepare First | **$1.50M** | D2 $0.25M, D6 $0.25M, D3 $0.25M |
| Aysén | Prepare First | **$1.75M** | D2 $0.50M, D5 $0.50M, D10 $0.25M |
| Los Ríos | Prepare First | **$1.00M** | D3 $0.25M, D10 $0.25M, D4 $0.25M |
| O'Higgins | Prepare First | **$1.00M** | D1 $0.25M, D3 $0.25M, D5 $0.25M |
| Cross-cutting: national predictive-monitoring & data-governance pilot | Cross-cutting | **$1.50M** | Sensor upkeep $0.50M; data platform & Tableau operations $0.40M; privacy & model-risk audit $0.30M; analytics staffing & data stewards $0.30M |

The distribution is intentionally asymmetric: $30.25M (67%) to the five Scale Now regions to accelerate demand while protecting capacity, $13.25M (29%) to the eight Prepare First regions for foundational work, and $1.50M (3%) to the cross-cutting monitoring and data-governance pilot funded from the governance (D10) lines of five regions; the Wait tier receives nothing. Summed across all regions, the largest dimension commitments are D2 Natural Resources & Sustainability ($7.75M), D4 Infrastructure ($7.25M), D5 Mobility ($7.25M) and D9 Promotion ($5.25M); promotion trails infrastructure and enabling factors because RfP 1's evidence found enabling conditions and digital capacity, not natural resources or advertising, drive tourism demand (Figure 4). Digital readiness itself is not a budget line: it is an outcome of enabling investment and private-sector activity, not something the Ministry buys directly, so the closest direct lever is the USD 1.5M monitoring and data-governance pilot below. The larger D2 and D4/D5 lines fund the carrying-capacity and mobility work Scale Now regions need to absorb demand that digital readiness is already pulling in; without it, digitally-driven demand converts into crowding and service failure, not durable revenue.

**Figure 4. Proposed USD 45M allocation by region and investment bucket**



## Implementation and governance

| Timing | Action | Accountability |
|---|---|---|
| 0-90 days | Operationalise the four Tableau dashboards; publish the data dictionary, quality register and privacy thresholds; set baseline RMSE and tier definitions. | Analytics lead + regional data stewards |
| Quarter 2 | Launch Scale Now campaigns with capacity safeguards; fund Prepare First gap projects; start monthly DB4 signal review and the champion-challenger forecast. | Tourism promotion + infrastructure + regional authorities |
| Quarter 3 | Compare realised demand with forecasts and the naive floor; audit crowding-alert false positives and negatives; first model-risk report. | Model-risk / analytics committee |
| Quarter 4 | Re-rank regions with refreshed data; publish a performance summary; reallocate uncommitted funds if a Prepare First region clears its bottleneck. | Vice Ministry investment committee |

### Decision rules

- Do not promote a region only because it has a high asset score. The region must also have enough supporting capacity or a funded plan to improve it.
- Do not make decisions based on only 1 digital signal. Always confirm the information using at least 2 different data sources when decisions affect public funding, public access, or safety.
- Keep the seasonal naive forecast as the main forecasting method. The regression model should replace it only for individual series after it performs better in 2 consecutive holdout tests. If it no longer performs better, it should return to the seasonal naive method.
- Review the Scale, Prepare, and Wait tiers every year and immediately after major structural changes, such as route changes, natural disasters, or long-term capacity constraints.

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